

Explainable Artificial Intelligence in Portfolio Management: A Literature Review

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Abstract—The use of advanced artificial intelligence models for portfolio management is growing as they provide effective strategies for dynamic markets where traditional approaches often fail. However, the black-box nature of these models hinders their adoption due to a lack of transparency for investors and regulators. This study reviews consolidated recent research on explainable artificial intelligence (XAI) as the solution to this problem. The study covers XAI applications in core portfolio tasks, including asset price prediction and asset allocation across various models, including deep reinforcement learning and hybrid framework. The findings show that XAI tools reveal the key features that drive the models' decisions, which is the essential step in transforming AI from an unexplainable system into a trusted and audible tool for modern investment.

Keywords—Explainable Artificial Intelligence, Price prediction, Portfolio Management, Model Interpretability, Agnostic Model, Post-Hoc analysis

I. INTRODUCTION

The advancement of Machine Learning (ML) algorithms and Artificial Intelligence (AI), particularly deep learning and reinforcement learning, has transformed data-driven modeling and decision-making across domains. Finance is one of the fields that also adopted these tools, as they could enhance the models to capture the time-series data that demonstrates the noisy, dynamically evolving, and nonlinear patterns.

Within financial applications, portfolio management has become one of the most critical AI applications where the developed model needs to perform two involving tasks that are selecting assets to include in the portfolio and allocating weights among them to optimize the portfolio risk and return. However, with the dynamic nature of financial markets, these models need to adapt to changing the market regime and discovering hidden patterns in the market behaviors where the traditional methods, such as mean-variance optimization [1] and econometric forecasting, often struggle to provide robust and timely solutions. In contrast, the studies on AI-based methods have shown strong potential to overcome these limitations by learning directly from data to detect hidden patterns and adjusting dynamically to the changes of the market conditions [2] [3]. These advances highlight

the incremental integration of AI into portfolio management research and practice.

While AI has demonstrated remarkable success across many domains, they often function as black-box systems, where the reasoning behind their decisions is limited. This general problem becomes critical in finance, especially in portfolio management, where the reasoning behind asset selection and allocation must be transparent to ensure the investor trust and compliance with the regulatory requirements. To address these challenges, the explainable AI (XAI) has emerged as an important research field that aims to enhance the interpretability of AI-driven models while retaining their predictive power. Recent financial studies have applied XAI techniques to clarify the drivers of allocation decision using both widely used methods, LIME [4] and SHAP [5], and financial-specific methods tailored to portfolio management [2] [3] [6]. These studies indicate that XAI improves the transparency of AI driven portfolio strategies and provides new opportunities to implement advanced AI methods with practical financial decision-making. Given these developments, a comprehensive review of XAI in portfolio management is needed to consolidate current progress and identify future directions.

The objective of this paper is to review recent advances in explainable AI applied in the portfolio management studies. The review consolidates current research, highlights commonly used and domain-specific XAI approaches and identifies directions for future work for the trustworthy AI driven portfolio strategies.

The findings in this paper are based on a systematic literature review. The search was conducted on ScienceDirect, IEEE Xplore Digital Library and Google Scholar for research papers published between 2016 and 2025. The query combined terms such as ("Explainable Artificial Intelligence" OR "XAI" OR "interpretable machine learning") AND ("portfolio management" OR "asset allocation" OR "asset price prediction"). The initial results were then filtered to include only peer-reviewed articles that present specific XAI techniques and evaluate their application in portfolio management tasks.

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II. EXPLAINABLE ARTIFICIAL INTELLIGENCE METHODS

The field of XAI has emerged to address the black-box nature of the modern AI models. The objective of XAI is to make the model predictions understandable to users while maintaining high predictive accuracy. Among the most widely adopted techniques in XAI are Local Interpretable Model-agnostic Explanation (LIME) and SHapley Additive exPlanation (SHAP). Both are designed to interpret the predictions of a machine learning model without needing to access or make assumptions about its internal processes, or a property known as model-agnostic. This flexibility has made them suitable for application across a wide variety of tasks including financial forecasting and portfolio management.

One of the earliest and influential contributions in XAI is LIME, proposed by Rebeiro et al. [4] [7]. LIME explains individual predictions by generating small variations of the input data around the instance and then training a simple surrogate model, for example linear regression, that approximates the behavior of the complex machine learning model. This obtained surrogate model's coefficients which reveal features importance on the predictions that human can interpret. Despite its influence, LIME has notable limitations on the unstable explanation due to the dependency on new local sample generated [4], and the method becomes computationally demanding when applied to large datasets [7]. To extend LIME for global explanations, Rebeiro et al. [4] also proposed Submodular Pick LIME or SP-LIME, which selects a representative set of instances to provide global understanding of model. However, SP-LIME has not been widely adopted because it still inherits the high computational cost of LIME.

Another widely used XAI method is SHAP, introduced by Lundberg and Lee [5]. SHAP quantifies a feature importance as a conditional expected value of the model output by comparing predictions when a feature is used in the model and when it is excluded. SHAP was developed based on three mathematical properties: local accuracy, missingness, and consistency. The local accuracy property requires that the sum of feature attributions plus the base value must be equal to the output of the original model. The missingness property guarantees that unused features in the model are assigned to zero value. The consistency stipulates that if a feature has a greater contribution to a model, its importance will not decrease. This final property is crucial for validating that explanations are reliable and reflect the true impact of the features. Additionally, SHAP provides not only local explanations for individual predictions but also allows the aggregation of the local results into global understanding of the machine learning model. This dual capability has made SHAP the most widely adopted XAI methods in practice.

However, both SHAP and LIME have several limitations. Their explanations may misrepresent a model's true internal logic, leading to unreliable explanation. The sensitivity of the changes in the data can affect the stability of the explanation output. Additionally, both provide only post-

hoc attributions rather than causal reasoning that can be difficult for non-technical stakeholders to interpret. From a regulators' perspective, these weaknesses raise concerns about whether explanations for complex AI models are reliable, reproducible, auditable, and legal defensible.

In addition to the general-purpose methods such as LIME and SHAP, researchers have proposed a set of financial XAI tools that are designed to interpret the outputs of complex models used in financial studies. Guan and Liu [2] proposed a two-part method to explain the decisions of deep reinforcement learning (DRL) agents. First, the integrated gradients are used to score which features are most important for specific actions. Then, a hindsight linear model approximates the agent's complex logic with a simple and interpretable model. This method allows portfolio rebalancing actions to be traced back to key market features. Similarly, Wang et al. [3] integrates attention mechanisms with sensitivity analysis. The attention part is used to determine which inputs that the model prioritize during decision-making, while sensitivity analysis evaluates how the model's predictions react to changes in those inputs. Together, this method could provide a complementary and holistic view of the model's behavior. Another category of methods is the post-hoc policy analysis method [8] which explains reinforcement learning models once the training is completed. This is achieved by extracting simplified decision rules or mapping state-action value back to input features to make the decision logic more interpretable. Although this method is good for generating insights, its explanations are just approximations and may fail to capture the full complexity of the original policy. Another framework enhances SHAP-based explanations with two additional layers [9]. First, it uses SHAP to determine feature importance and then clusters these attributions to find consistent decision patterns. On top of that, this method adds uncertainty estimation to assess the model's confidence. This combined technique produces a robust explanation that covers the logic of decision and reliability in the volatile market environment. The evolution from model-agnostic techniques to specialized financial frameworks shows how XAI has evolved, building trust and transparency needed to apply these models to tasks like asset prediction and portfolio allocation.

III. ASSET PRICE PREDICTION AND EXPLAINABLE ARTIFICIAL INTELLIGENCE

The first task in portfolio management is asset selection, in which the goal is to identify which assets should be included in a portfolio. A wide range of prediction models, from machine learning to deep learning, are applied to forecast the asset prices, market movements, and sentiment. Although these models could outperform traditional statistical methods, they lack transparency due to their black-box nature. Therefore, XAI methods are essential for revealing which features drive the model's predictions. As shown in Table I, several recent studies have applied tools like LIME and SHAP to achieve this.

The application of XAI also extends to classical machine learning and hybrid models. Lin et al. [10] proposed a combined technical and sentiment analysis into machine learning classifiers. They used LIME to enhance interpretability which revealed which financial indicators or sentiment features were most influential for model's asset recommendations. Similarly, Tanveer et al. [11] used LIME and SHAP to interpret a complex hybrid model that combined Convolution Neural Network (CNN) and Bidirectional Long Short Term Memory (Bi-LSTM). Their work revealed that XAI tools offer a more comprehensive view of what drives the model's predictions. To further ground such hybrid models in established financial theory, Rajan [6] first used ARIMAX model to capture linear relationships and then use its extracted features as additional feature for a CNN-LSTM model. The use of SHAP on the final model revealed that trading volume, recent price changes, and volatility were among the most influential factors that drive the forecasts.

TABLE I
XAI METHODS IN PRICE PREDICTION

Ref.	Prediction Method	XAI Tool(s)
[6]	ARIMAX + CNN-LSTM hybrid model	SHAP
[10]	ML classifiers with technical & sentiment features	LIME
[11]	CNN + Bi-LSTM for stock forecasting	LIME, SHAP
[12]	Deep neural network for trend forecasting	LIME, SHAP
[13]	Sequential models (Conv1D, LSTM)	LIME, SHAP

Deep learning models have also been paired with XAI to improve interpretability in stock price prediction. Muhammad et al. [12] developed a deep neural network for market trend predictions and applied both LIME and SHAP to interpret model outputs. The analysis revealed that the technical analysis indicators and market features such as volatility and momentum, were primary contributors to the forecasting, which makes the models' asset recommendation more transparent. Similarly, Sasikumar and Raj [13] studied sequential forecasting using various deep learning architectures, including Conv1D and LSTM. The study found that using both approaches together provide a more comprehensive view of the patterns learned by models. This was a key step in showing the importance of time-lagged variables and explaining the logic behind asset price forecasts.

In summary, these studies demonstrate how XAI methods enhance prediction tasks by clarifying the key features that drive asset selection decisions. This not only enhances transparency and reliability of AI-driven forecasting but also builds confidence in decision-making in asset selection process. This strengthens the foundation for constructing interpretable portfolio strategies.

IV. ASSET ALLOCATION AND EXPLAINABLE ARTIFICIAL INTELLIGENCE

The second task in portfolio management is asset allocation that the weight for each asset in portfolio will be deter-

mined. The classical approach for portfolio management was introduced by Markowitz [1], also known as mean-variance optimization (MVO) method. This framework constructed a portfolio by solving the optimization problem that balances the expected return against its risk measured by variance of its returns. However, the assumption of this method relies on the normally distributed returns and stable market conditions which often fail in current markets. The observation is that the distribution for the asset returns tends to demonstrate the fat-tailed distribution, which limits the benefits of MVO method [14]. This limitation is particularly evident in high volatile and non-stationary market such as the technology stocks or cryptocurrencies [8]. These limitations have driven a shift of portfolio management toward dynamic and data-driven methods such as Deep Reinforcement Learning (DRL). While DRL-based allocation can adapt flexibly to dynamic market conditions [2] [3] [15], its black-box nature still limits their transparency that makes interpretation by users difficult. This motivates the integration of XAI to make allocation more interpretable. As summarized in Table II, the following review details how a range of XAI techniques have been applied to this challenge.

TABLE II
XAI METHODS IN ASSET ALLOCATION

Ref.	Allocation Method	XAI Tool(s)
[2]	DRL and ML models (Random Forest, SVM)	Integrated Gradients
[3]	DRL with attention networks	Attention Mechanisms; Sensitivity Analysis
[8]	DRL (PPO algorithm)	SHAP; LIME
[9]	Two-level DRL architecture	SHAP; Attribution Clustering
[15]	DRL (A2C agent with LSTM)	Attention Mechanisms
[16]	Hybrid (Dynamic Mean-Variance + ML)	SHAP
[17]	Hybrid (XGBoost for HRP vs. ERC)	SHAP
[18]	Hybrid (Clustering + Black-Litterman)	Structural Explainability

To address this black-box nature of DRL, the XAI techniques have been applied to reveal how the models use input features to make allocation decisions. Guan and Liu [2] used integrated gradient to measure the features 'contribution for their developed portfolio strategies based on machine learning algorithms, such as random forest and SVM, and DRL. The XAI analysis revealed the source of DRL agent's strategy was aligned with an optimal long-term growth of wealth while the machine learning strategies focused on the short-term prediction. Similarly, other studies embed interpretability directly into DRL architecture using attention mechanisms to dynamically measure feature importance. Wang et al. [3] used this method in the proposed algorithm with sensitivity analysis of the attention weights. The study found that the agent learned to prioritize features related to long-term growth and low volatility. Similarly, Cortes et al [15] applied the attention layers with the proposed framework combining reinforcement learning algorithm with LSTM. The XAI analysis revealed that the model learns to assign the weight features of each asset differently when constructing portfolio as the importance of the inputs like moving averages and trading volume varied

significantly from one stock to another. In contrast, another approach uses post-hoc analysis to explain trained agent. de-la-Rica-Escudero et al. [8] applied SHAP and LIME to the DRL agent trained using the Proximal Policy Optimization (PPO) algorithm. The XAI analysis revealed the importance of price features (Open, High, Low, Close) on individual stock allocations. In a related study, Lina et al. [9] used SHAP to explain the proposed two-level DRL architecture for portfolio management. SHAP analysis provided a global explanation on the agent's decision-making, showing the model assigned greater importance to market uncertainty and asset correlation rather than to traditional trading signals.

The hybrid frameworks that combined machine learning with traditional financial theory offer another way to create explainable models. Zhang et al. [16] proposed the dynamic asset allocation based on market regimes. The model's explainability directly comes from its structure that first used hierarchical clustering on macroeconomic and market data to identify market regimes. The expected return and covariance calculated from each specific regime are then used to form the investor's view within the Black-Litterman portfolio optimization model. This structure provides transparency since the final asset allocation is directly determined by the data-driven and interpretable market regime.

The implementations of XAI are further demonstrated in specific asset allocation challenges. In the high volatility market, Babaei et al. [17] used SHAP to explain a machine learning model trained on the outputs of a dynamic mean-variance portfolio. SHAP enhanced transparency for investors by revealing which assets were the most influential driver of the automated allocation process. For the broader challenge of constructing diversified portfolios, Jaeger et al. [18] trained XGBoost model to predict the performance differences between Hierarchical Risk Parity (HRP) portfolio and Equal Risk Contribution (ERC). The explanatory was done by SHAP that revealed the HRP's outperformance was primarily driven by better handling of the downside risk of individual assets in portfolio.

Collectively, these studies demonstrate that XAI enhances portfolio allocation by making the underlying models interpretable. The key features that drive asset allocation decisions were explained using XAI tools such as integrated gradients, attention mechanisms, and post-hoc analysis. This move toward transparency allows AI-driven finance to become a trusted and auditable tool for modern investment, instead of just a high-performance black-box method.

V. CONCLUSIONS

This study's analysis shows the growing importance of explainability for artificial intelligence in financial asset management. The limitations of the traditional approaches, such as Mean-Variance Optimization have motivated the use of more powerful machine learning methods including deep reinforcement learning algorithms that have been applied for their ability to adapt to dynamic financial markets. However,

the main obstacle to their widespread use is a lack of interpretability and transparency. To solve this issue, the studies in this review apply a range of explainable artificial intelligence methods, from post-hoc tools, such as LIME and SHAP, to embed techniques like attention layers. These approaches succeed in showing the logic behind both forecasting and weight allocation by highlighting the influential features. Although this research is still emerging, the work to make these models interpretable is essential for allowing AI-driven finance to move from a powerful but black-box system into a dependable and practical tool for investors.

Futures studies could focus on several key areas to advance the field. One important direction is the development of novel hybrid XAI framework. For example, a framework that could integrate the strengths of different techniques, combining the feature-level insight from integrated gradients with the post-hoc analysis of SHAP to create more robust explanations. Finally, further studies are needed to ensure these advanced explanations satisfy the transparency and auditability requirements of financial regulators.

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